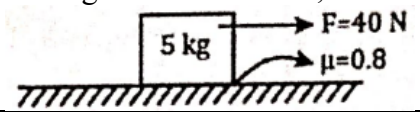
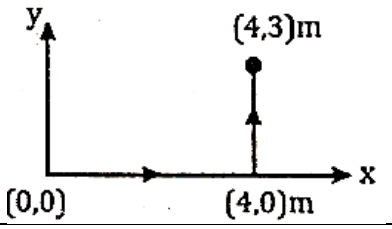
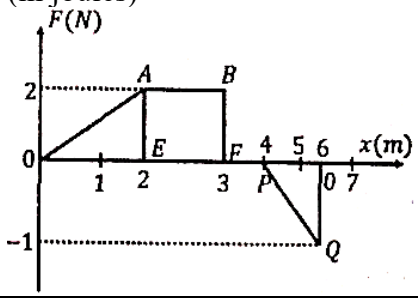
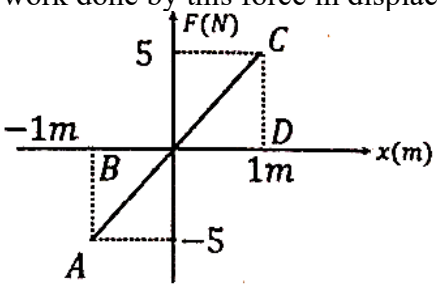
Question	A body moves a distance of 5m along straight line under the action of a force of 15 N. If work done by this force is 45J, the angle which the force makes with the direction of motion with the horizontal is
Type	multiple choice
Option	37^0
Option	30^0
Option	60^0
Option	53^0
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Work done when a force $\vec{F}=\left(\hat{i}+2\hat{j}+3\hat{k}\right)N$ acting on a particle displaces it from the point $r_1=\left(\hat{i}+\hat{j}+\hat{k}\right)$ to the point $r_2=\left(\hat{i}-\hat{j}+2\hat{k}\right)$
Type	multiple choice
Option	$-3J$
Option	$7J$
Option	$-1J$
Option	$1J$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A constant force $\vec{F}=2\hat{i}-3\hat{j}+\hat{k}$ is applied on a ball to displace it from $\vec{r}_1=\hat{i}+\hat{j}$ to $\vec{r}_2=4\hat{i}+3\hat{j}$, which of the following statements is true?
Type	multiple choice
Option	A force and displacement are perpendicular to each other and work done by force is equal to zero
Option	Force and displacement are parallel to each other and work done by force is equal to zero
Option	Force and displacement are anti parallel to each other and work done by force is equal to zero
Option	Force and displacement are perpendicular to each other and work done by force is not equal to zero
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A force of 5N, making an angle θ with the horizontal acting on an object and displaces it by 0.4m along the horizontal direction. If the work done by the force is 1J, then the horizontal component of the force is
Type	multiple choice
Option	1.5 N
Option	2.5 N
Option	3.5 N
Option	4.5 N
Answer	2
Solution	
Positive Marks	4
Negative Marks	1

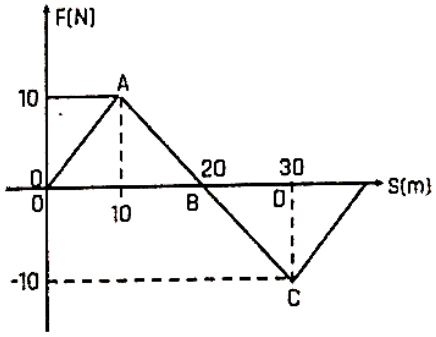
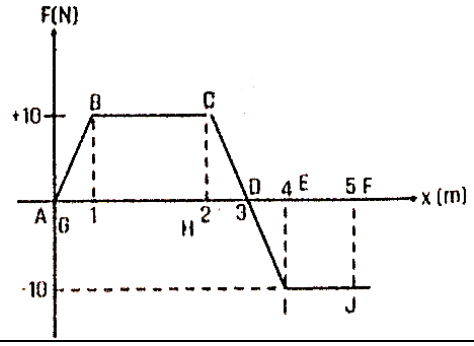
Question	A body of mass 2kg is thrown vertically upwards with an initial speed of 10m/s. The work done by the force of gravity during the time the body reaches its maximum height. (Consider $g = 10m/s^2$)
Type	multiple choice
Option	100J
Option	0J
Option	-100J
Option	20J
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A body having a mass of 100 gm is allowed to fall freely (from a very large height) under the action of gravity. Work done by gravity during first 10 s of journey is (Take g = 10 m/s²)
Type	multiple choice
Option	500000 J
Option	50000 J
Option	5000 J
Option	500 J
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In a figure shown below, the work done by force of friction on the block is 
Type	multiple choice
Option	Zero
Option	Positive
Option	Negative
Option	cannot be determined
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A force $\vec{F} = (3t+4)\hat{J}$ N acts on a particle moving in x - y plane. Starting from origin, the particle first goes along x-axis to the point (4 m, 0) and then parallel to the y-axis to the point (4 m, 3 m). The total work done by the force on the particle is 
Type	multiple choice
Option	+12 J
Option	-6 J
Option	+24 J
Option	-12 J
Answer	3

Solution	
Positive Marks	4
Negative Marks	1
Question	A force $F = 10 + 2y$ acts on a particle in y-direction, where F is in Newtons and y is in meters. Work done by the force to move the particle form origin to 2 m along +y direction is
Type	multiple choice
Option	24 J
Option	32 J
Option	48 J
Option	12 J
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A force $F = 5 + 2x + 3x^2$ acts on a particle in x direction where F is in Newton and x is in meter. Find the work done by this force during a displacement from x = 1 m to x = 2 m.
Type	multiple choice
Option	15 J
Option	12 J
Option	32 J
Option	-15 J
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle moves under the effect of force $F = kx^2$. Find work done by force when particle displaces from $x = 0m$ to $x = 4m.$
Type	multiple choice
Option	$\frac{8k}{3}$
Option	$\frac{32k}{3}$
Option	$\frac{64k}{3}$
Option	$\frac{128k}{3}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	An object is displaced from point A(2,3,4) m to a point B(1,2,3)m under the influence of a force $\vec{F} = 2\hat{i} + 3y^2\hat{j} + 5\hat{k}.$ The work done by this force is
Type	multiple choice
Option	26 J
Option	- 80 J
Option	- 26 J
Option	34 J
Answer	3
Solution	

Positive Marks	4
Negative Marks	1
Question	A child applies a force F parallel to the x-axis to a block moving on a horizontal surface. The x -coordinate of the block as shown in figure. The work done by the force F when the block moves from $x=0$ to $x=7m$ and $x=3m$ to $4m$ are (in joules) 
Type	multiple choice
Option	$-3,0$
Option	$0,0$
Option	$3,0$
Option	$-3,3$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A force F acting on a particle varies with position x as shown in figure. The work done by this force in displacing the particle from x = -1m to 1m. 
Type	multiple choice
Option	+20 J
Option	0J
Option	-20 J
Option	+10 J
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statements regarding conservative forces is NOT correct?
Type	multiple choice
Option	Total energy remains constant
Option	Work done in round trip is zero
Option	Work done is dependent on path
Option	Work done is completely recoverable
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A light and a heavy body have equal momenta. Which one has greater K.E

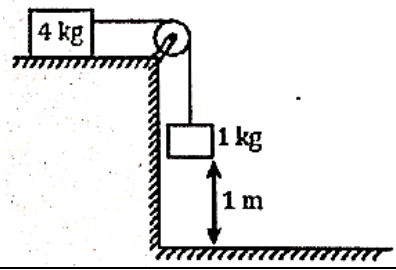
Type	multiple choice
Option	The light body
Option	The heavy body
Option	The K.E. are equal
Option	Data is incomplete
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	An athlete of mass 50 kg in the Olympic games covers a distance of 100m in 10s. His kinetic energy can be estimated to be in the range (Assume velocity is constant)
Type	multiple choice
Option	3000 J
Option	2500 J
Option	5,000 J
Option	4000 J
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball of mass 2kg and another of mass 4kg are dropped together from a 60 feet tall building. After a fall of 30 feet each towards earth, their respective kinetic energies will be in the ratio of
Type	multiple choice
Option	$\sqrt{2}:1$
Option	$1:4$
Option	$1:2$
Option	$1:\sqrt{2}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The kinetic energy of a body of mass 3kg and momentum 2 Ns is
Type	multiple choice
Option	$1J$
Option	$\frac{2}{3}J$
Option	$\frac{3}{2}J$
Option	$4J$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two bodies of masses 2m and m have their K.E. in the ratio 8:1, then their ratio of momentum is
Type	multiple choice
Option	$1:1$
Option	$2:1$
Option	$4:1$

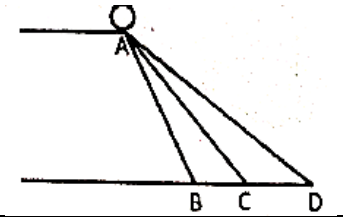
Option	$8:1$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If a man increase his speed by 2m/s, his K.E, is doubled, the original speed of the man is
Type	multiple choice
Option	$\left(1+2\sqrt{2}\right)m/s$
Option	$4m/s$
Option	$\left(2+2\sqrt{2}\right)m/s$
Option	$\left(1+\sqrt{2}\right)m/s$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball of mass 2 kg is projected horizontally with a speed of 3m/s from top of a tower. The kinetic energy of the ball after 0.4 s is
Type	multiple choice
Option	$25J$
Option	$16J$
Option	$9J$
Option	$8J$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A body moving with velocity has momentum and kinetic energy numerically equal. What is the value of v?
Type	multiple choice
Option	$2m/s$
Option	$\sqrt{2}m/s$
Option	$1m/s$
Option	$0.2m/s$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of mass 1 kg is thrown upwards with a velocity 10 m/s. It momentarily comes to rest after attaining a height of 4 m. How much energy is lost due to air friction? ($g = 10\ m/s^2$)
Type	multiple choice
Option	$40J$
Option	$50J$
Option	$10J$
Option	$100J$
Answer	3

Solution	
Positive Marks	4
Negative Marks	1
Question	A man applies a constant force of 5 N on an object of mass 2 kg and displaces it by 5 m. The object was initially at rest. The final velocity of object is
Type	multiple choice
Option	$10 m/s$
Option	$5 m/s$
Option	$25 m/s$
Option	$15 m/s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Force versus displacement curve is shown in the diagram. Find the change in kinetic energy by the force at the end of 30 m. 
Type	multiple choice
Option	$25 J$
Option	$50 J$
Option	$75 J$
Option	$100 J$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A position dependent force F is acting on a particle and its force position curve is shown in the figure. Change in kinetic energy of the particle, when its displacement is from 0 to 5 m is 
Type	multiple choice
Option	$5 J$
Option	$10 J$
Option	$15 J$
Option	$20 J$

Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	An object of mass 50 kg is raised to a height of 10 m above the ground. If the object is allowed to free fall. What is kinetic energy, when it is half way down? $\left(g = 10 m/s^2\right)$
Type	multiple choice
Option	$1250 J$
Option	$2500 J$
Option	$3750 J$
Option	$5000 J$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A bullet of mass 50 g leaves a rifle at an initial velocity of 2000 m/s and strickes the earth at the same level with a velocity of 1000 m/s. The work done in overcoming the resistance of air will be
Type	multiple choice
Option	$750\ J$
Option	$7.5 \times 10^3\ J$
Option	$4500 J$
Option	$7.5 \times 10^4 J$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A bullet of mass 20 g leaves a riffle at an initial speed 100 m/s and strikes a target at the same level with speed v m/s. If the amount of work done by the resistance of air is 751, the value of v will be
Type	multiple choice
Option	$100 m/s$
Option	$25 m/s$
Option	$75 m/s$
Option	$50 m/s$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Consider the following statements- 1. Gravitational force is an example of a conservative force 2. Frictional forces is an example of non-conservative force Which of the following statements is Correct?
Type	multiple choice
Option	1 only
Option	2 only
Option	Both 1 and 2
Option	Neither 1 nor 2
Answer	3

Solution	
Positive Marks	4
Negative Marks	1
Question	Potential energy of a particle is given by $U=\left(2-3x-4x^2\right)J$ where $'x'$ is in $'m'$. Find the magnitude of force at $x=4m,$ if particle is free to move along $x-$direction.
Type	multiple choice
Option	29 N
Option	35 N
Option	16 N
Option	18 N
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Potential energy of a particle moving along x-axis is given by $U=\left(\frac{x^3}{3}-4x+6\right)$. Here, U is in joules and x in meters. Find the position of stable and unstable equilibriums, respectively.
Type	multiple choice
Option	$2m,-2m$
Option	$-2m,2m$
Option	$0m,0m$
Option	$1m,-1m$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A bob of a simple pendulum, which is initially at rest at its mean position, is now raised to a vertical height of 1.8 cm and released from rest. The velocity of the bob at the mean position is
Type	multiple choice
Option	$6m/s$
Option	$36m/s$
Option	$0.6m/s$
Option	$6m/s$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle of mass 10 kg is thrown at angle of 37° with horizontal with an initial velocity of 5m/s. The kinetic energy of the projectile when it just reaches the ground will be
Type	multiple choice
Option	250 J
Option	125 J
Option	100 J
Option	80 J
Answer	2
Solution	

Positive Marks	4
Negative Marks	1
Question	A block is placed on the top of a plane inclined at 30° with horizontal. The length of the plane is 10m. The block slides down the plane and reaches the bottom. If the surface is frictionless, the speed of the block at the bottom of the inclined plane is
Type	multiple choice
Option	$10\,m/s$
Option	$10\sqrt{2}\,m/s$
Option	$5\,m/s$
Option	$5\sqrt{2}\,m/s$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of mass m dropped from a certain height strikes a light vertical fixed spring of stiffness k. The height of its fall before touching the spring if the maximum compression of the spring is equal to $\frac{3mg}{k}$ is
Type	multiple choice
Option	$\frac{3mg}{2k}$
Option	$\frac{2mg}{k}$
Option	$\frac{3mg}{4k}$
Option	$\frac{mg}{4k}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A 4kg block is on a smooth horizontal. Table. The block is connected to a second block of mass 1kg by a massless flexible tight cord that passes over a frictionless pulley. The 1 kg block is 1 m above the floor. The two blocks are released from rest. With what speed (in m/s) does the 1kg block hit the ground? 
Type	multiple choice
Option	3 m/s
Option	4 m/s
Option	2 m/s
Option	5 m/s
Answer	3
Solution	

Positive Marks	4
Negative Marks	1
Question	A ball is dropped along smooth inclined surfaces AB, AC and AD. The velocity acquired at B, C D be u_B, u_C and u_D respectively. 
Type	multiple choice
Option	$u_B > u_C > u_D$
Option	$u_B < u_C < u_D$
Option	$u_B > u_C < u_D$
Option	$u_B = u_C = u_D$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The potential energy of a particle that is free to move along (x - axis) is given by $U = \left(x^3 - \frac{x^2}{2}\right) J$, where x is in $'m'$. The particle is initially at $x = 0$. Find the magnitude of force at $x = 2m$.
Type	multiple choice
Option	$2\,N$
Option	$4\,N$
Option	$8\,N$
Option	$10\,N$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A block of mass m is attached to one end of a string. The other end is fixed. The block is rotated in vertical circle of radius r. What is the tension at lowest point if velocity at that position is v?
Type	multiple choice
Option	$\frac{mv^2}{r}$
Option	$\frac{mv^2}{r} - mg$
Option	$\frac{mv^2}{r} + mg$
Option	mg
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of mass 0.4kg is whirled in a vertical circle making 2rev/sec. If the radius of the circle is 1.2 m, then tension in the string when the body is at the top

	of the circle, is
Type	multiple choice
Option	52 N
Option	50 N
Option	60 N
Option	72 N
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A bucket full of water is revolved in vertical circle of radius 2m. What should be the maximum approx. time-period of revolution so that the water doesn't fall off the bucket? (Assume w is constant)
Type	multiple choice
Option	1 sec
Option	7 sec
Option	3 sec
Option	5 sec
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A block of mass m at the end of a string is whirled round in a vertical circle of radius R. The critical speed of the block at the top of its swing below which the string would slacken before the block reaches the top is
Type	multiple choice
Option	Rg
Option	$(Rg)^2$
Option	R/g
Option	$\sqrt{Rg}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle of mass 200 g, is whirled into a vertical circle of radius 80 cm using massless string. The speed of the particle when the string makes an angle of the 60° with the vertical line is $1.5ms^{-1}$. The tension in the string at this position is
Type	multiple choice
Option	4 N
Option	1.56 N
Option	2 N
Option	3 N
Answer	2
Solution	
Positive Marks	4
Negative Marks	1